

Circles and Parabolas

ANSWER KEY



Circles

- 1 Write the standard-form equation of the circle with center $(3, -2)$ and radius 5.
 $(x - 3)^2 + (y + 2)^2 = 25$.
 - 2 Consider the circle $x^2 + y^2 - 6x + 4y - 12 = 0$.
 - a Write the equation in standard form by completing the square. $(x - 3)^2 + (y + 2)^2 = 25$.
 - b State the center and radius. Center $(3, -2)$, radius 5.
 - 3 Determine whether the point $(6, 2)$ lies inside, on, or outside the circle $(x - 3)^2 + (y + 2)^2 = 25$.
 $(6 - 3)^2 + (2 + 2)^2 = 9 + 16 = 25$, so the point lies on the circle.
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Parabolas with vertex at the origin

- 4 For the parabola $y^2 = 8x$, find:
 - a the focus $(2, 0)$
 - b the directrix $x = -2$
 - c the direction of opening opens right
 - 5 A parabola has vertex at the origin, opens upward, and passes through the point $(4, 2)$.
 - a Find its equation in the form $x^2 = 4py$. $16 = 4p(2)$ gives $p = 2$, so $x^2 = 8y$.
 - b State the coordinates of the focus. $(0, 2)$.
 - 6 A satellite dish has a parabolic cross-section 2 m wide and 0.5 m deep. Taking the vertex at the origin opening upward, find the distance from the vertex to the focus (where the receiver is placed).
The edge point $(1, 0.5)$ lies on $x^2 = 4py$: $1 = 4p(0.5) = 2p$, so $p = 0.5$ m. The receiver sits 0.5 m above the vertex.
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Parabolas with vertex not at the origin

- 7 For the parabola $(y - 2)^2 = 12(x + 1)$, find:
 - a the vertex $(-1, 2)$
 - b the value of p $4p = 12$, so $p = 3$
 - c the focus $(2, 2)$
 - d the directrix $x = -4$
- 8 Write the equation of the parabola with vertex $(2, -3)$, opening downward, with $|p| = 1$.
Opening downward: $(x - 2)^2 = -4(y + 3)$.

- 9 Complete the square to write $y = x^2 - 6x + 5$ in vertex form, and state the vertex.
 $y = (x - 3)^2 - 4$. Vertex: $(3, -4)$.
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Mixed practice: circles and parabolas together

- 10 A circle and a parabola are given by $x^2 + y^2 = 25$ and $y = x^2 - 5$. Find their points of intersection.
Substitute $x^2 = y + 5$ into the circle: $(y + 5) + y^2 = 25$, so $y^2 + y - 20 = 0$, giving $(y + 5)(y - 4) = 0$:
 $y = -5$ or $y = 4$. For $y = -5$: $x^2 = 0$, point $(0, -5)$. For $y = 4$: $x^2 = 9$, points $(\pm 3, 4)$. Three intersection points: $(0, -5)$, $(3, 4)$, $(-3, 4)$.
- 11 A circular satellite orbit has the equation $x^2 + y^2 = (6400 + 500)^2$ (Earth radius plus altitude, in km). Find the radius of this orbit.
Radius = $6400 + 500 = 6900$ km.
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Graphing circles and parabolas

- 12 The parabola $y = x^2 - 2x - 3$ is shown. Use the graph to estimate its vertex and x-intercepts, then verify algebraically.
From the graph, x-intercepts near $x = -1, 3$ and vertex near $(1, -4)$. Algebraically: $y = (x - 1)^2 - 4$, vertex $(1, -4)$; setting $y = 0$: $(x - 1)^2 = 4 \Rightarrow x = -1, 3$ ✓.
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Tangent lines to circles

- 13 Find the equation of the tangent line to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$, using the fact that the tangent is perpendicular to the radius at that point.
Radius slope from origin to $(3, 4)$: $\frac{4}{3}$. Tangent slope (perpendicular): $-\frac{3}{4}$. Tangent line:
 $y - 4 = -\frac{3}{4}(x - 3)$, i.e. $y = -\frac{3}{4}x + \frac{25}{4}$.
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The reflective property of parabolas

- 14 Explain the reflective property of a parabola that makes it useful for satellite dishes and headlights, in terms of the focus.
Any ray traveling parallel to a parabola's axis of symmetry reflects off the interior surface and passes through the focus (or, conversely, rays originating at the focus reflect out parallel to the axis) — this is why satellite dishes concentrate incoming signals at the focus, and headlights placed at the focus project a parallel beam.
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Finding a circle through three points

- 15** Explain the general strategy for finding the equation of a circle passing through three given (non-collinear) points, without fully solving a specific example.

Substitute each of the three points into the general circle equation $x^2 + y^2 + Dx + Ey + F = 0$, producing a system of three linear equations in D, E, F . Solving this system gives the circle's equation, which can then be converted to standard form by completing the square.

Distance from a point to the focus (parabola definition)

- 16** State the focus-directrix definition of a parabola, and use it to verify that the point $(4, 4)$ lies on $y^2 = 4x$ (focus $(1, 0)$, directrix $x = -1$).

A parabola is the set of points equidistant from the focus and directrix. At $(4, 4)$: distance to focus $(1, 0)$ is $\sqrt{(4-1)^2 + 4^2} = \sqrt{9+16} = 5$. Distance to directrix $x = -1$ is $4 - (-1) = 5$. Equal, confirming the point lies on the parabola.

Systems involving circles

- 17** Find the points of intersection of the circles $x^2 + y^2 = 25$ and $(x-6)^2 + y^2 = 25$.

Subtracting the equations: $x^2 - (x-6)^2 = 0 \Rightarrow x^2 - (x^2 - 12x + 36) = 0 \Rightarrow 12x - 36 = 0 \Rightarrow x = 3$.

Substituting into the first: $9 + y^2 = 25 \Rightarrow y = \pm 4$. Intersection points: $(3, 4)$ and $(3, -4)$.